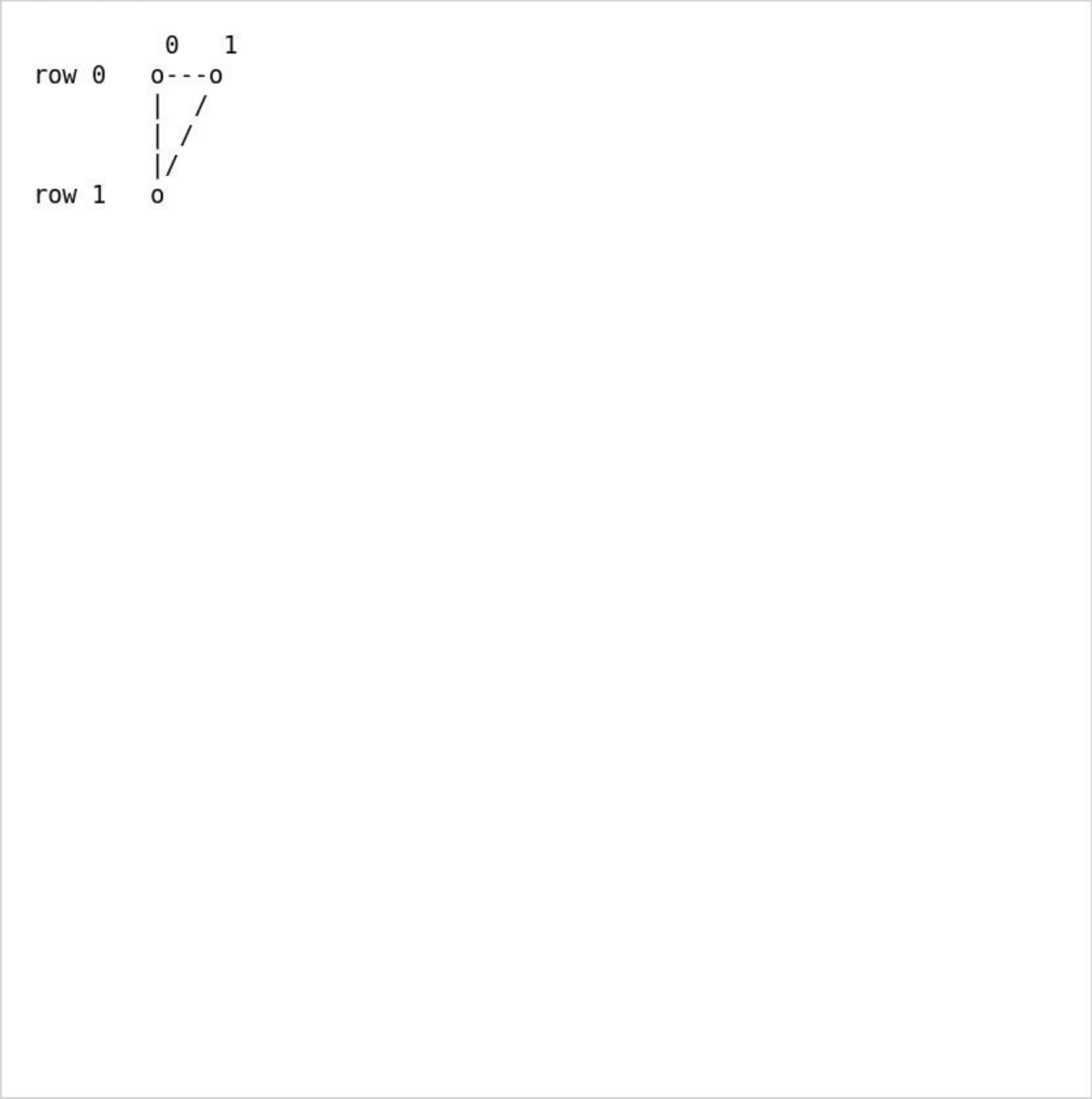


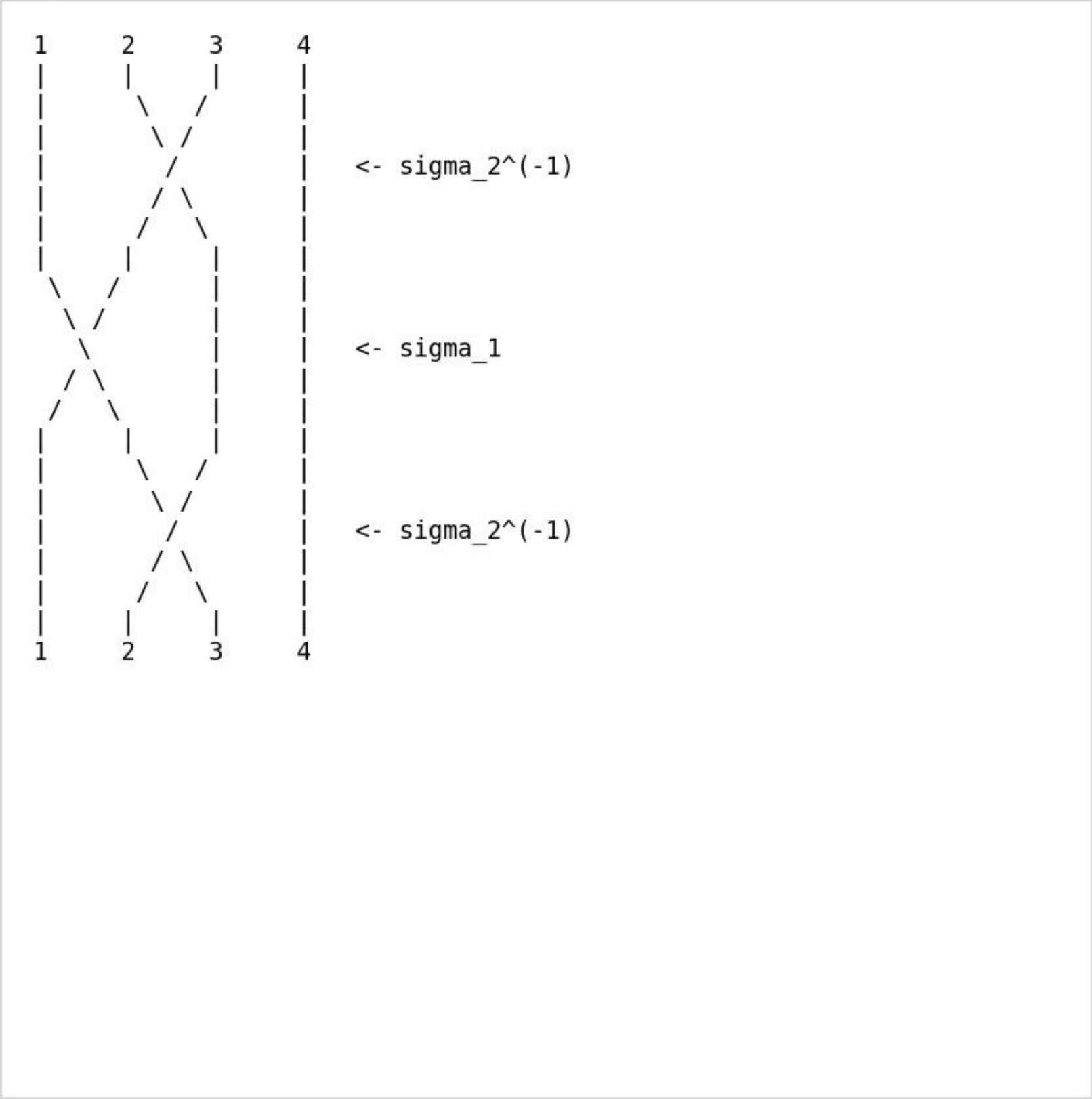
Example 1: Triangle — vertices (0,0), (0,1), (1,0)

Graph2Braid.py examples/example01_triangle.json

Input graph



Output braid



Example 2: Single non-vertical edge between (0,0) and (0,1)

Graph2Braid.py examples/example02_single_edge.json

Input graph

```

      0  1
row 0  0 - - 0

```

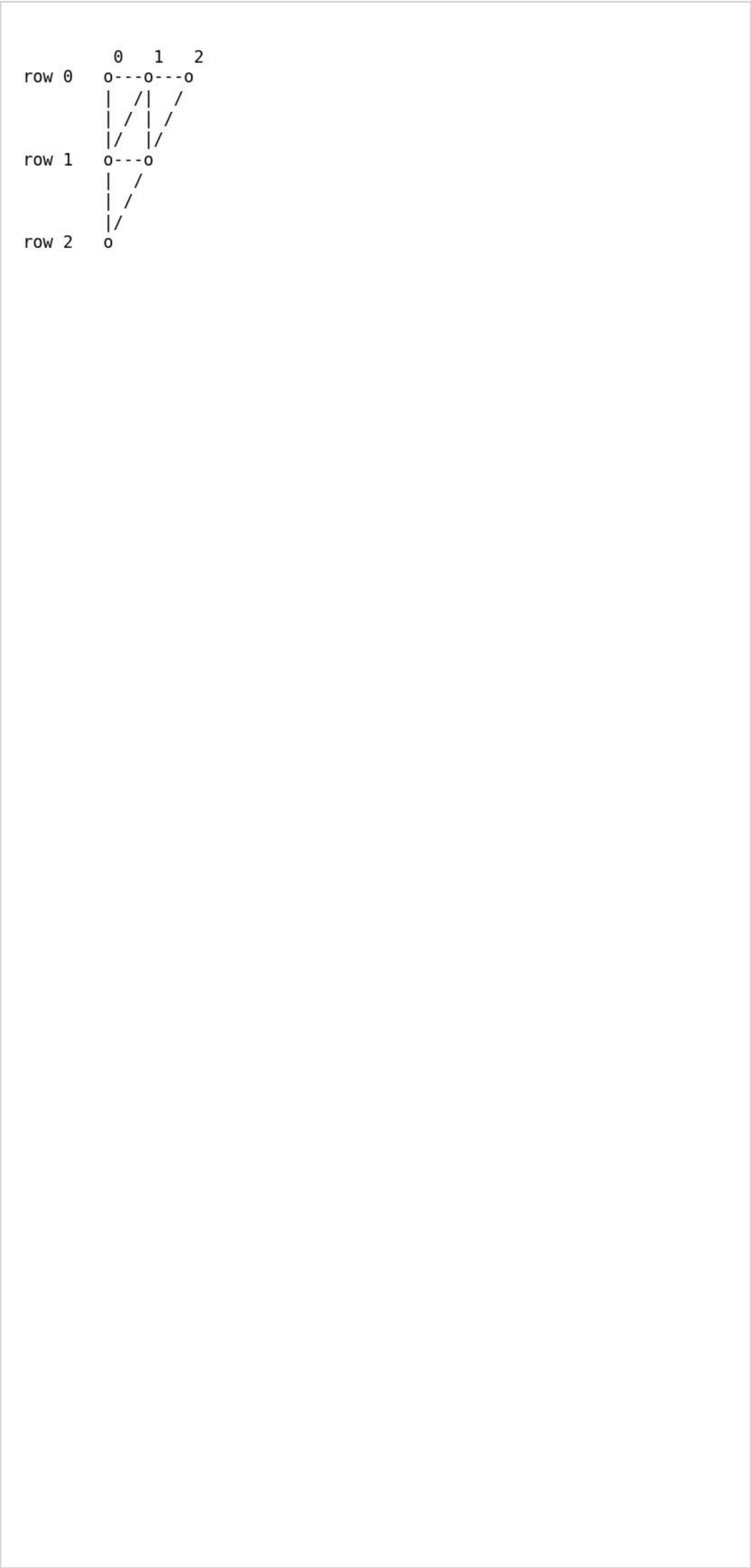
Output braid

$\leftarrow \sigma_2^{-1}$

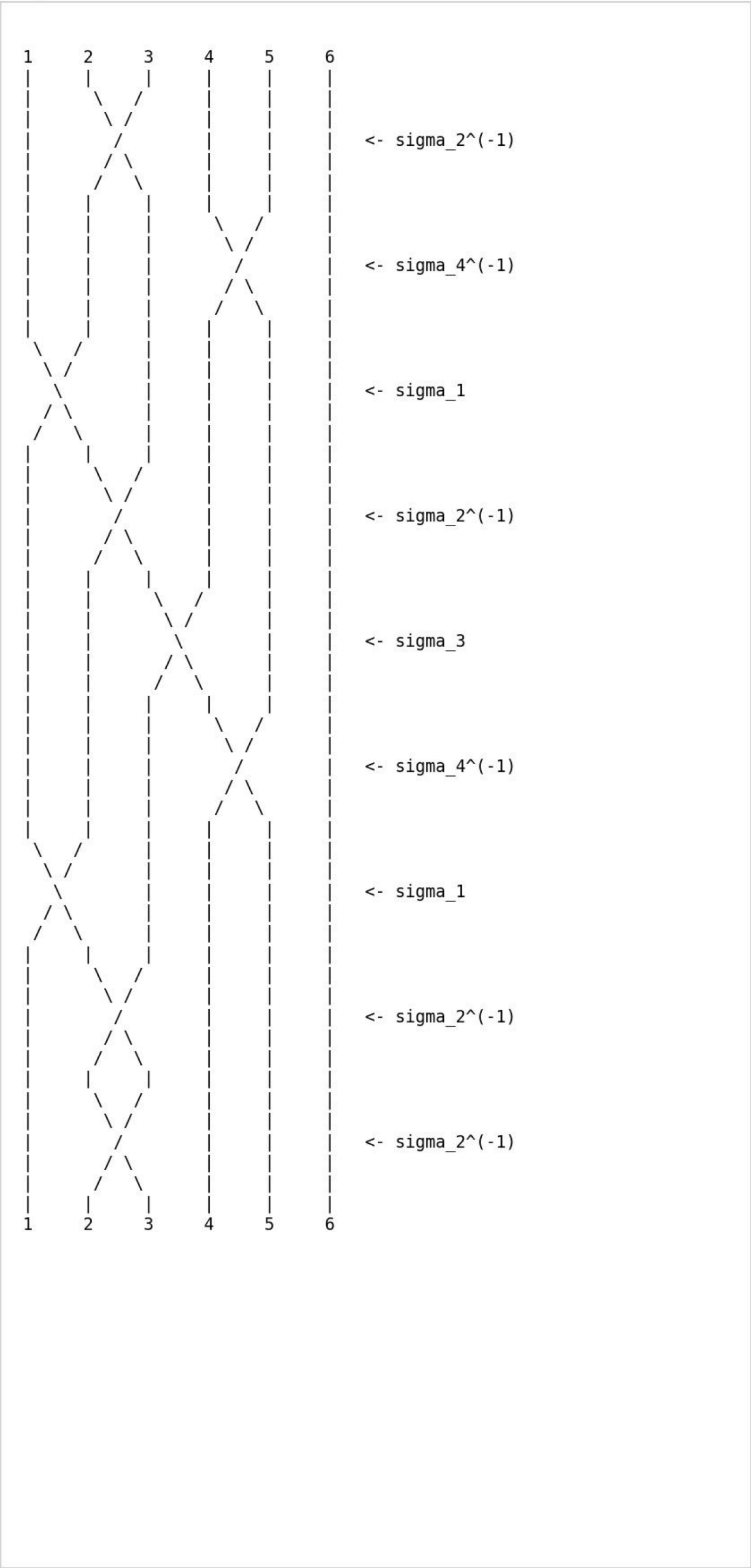
Example 3: 3-row staircase on a 3x3 grid

Graph2Braid.py examples/example03_staircase.json

Input graph



Output braid



Example 4: Single edge with weight 2 (two parallel edges between $(0,0)$ and $(0,1)$)

Graph2Braid.py examples/example04_double_edge.json

Input graph

```

      0  1
row 0  0 - - - 0

```

Output braid

Example 5: Single vertex (0,0) — trivial graph

Graph2Braid.py examples/example05_single_vertex.json

Input graph

		0
row 0	o	

Output braid

1	2
1	2

Example 6: Four vertices in column 0 — no non-vertical edges

Graph2Braid.py examples/example06_path_col0.json

Input graph

	0
row 0	o
row 1	o
row 2	o
row 3	o

Output braid

1	2	
\	/	
		<- sigma_1
/	\	
\	/	
		<- sigma_1
/	\	
1	2	

Example 7: Two vertices, one horizontal edge

Graph2Braid.py examples/example07_horizontal.json

Input graph

```

      0  1
row 0  0 - - 0

```

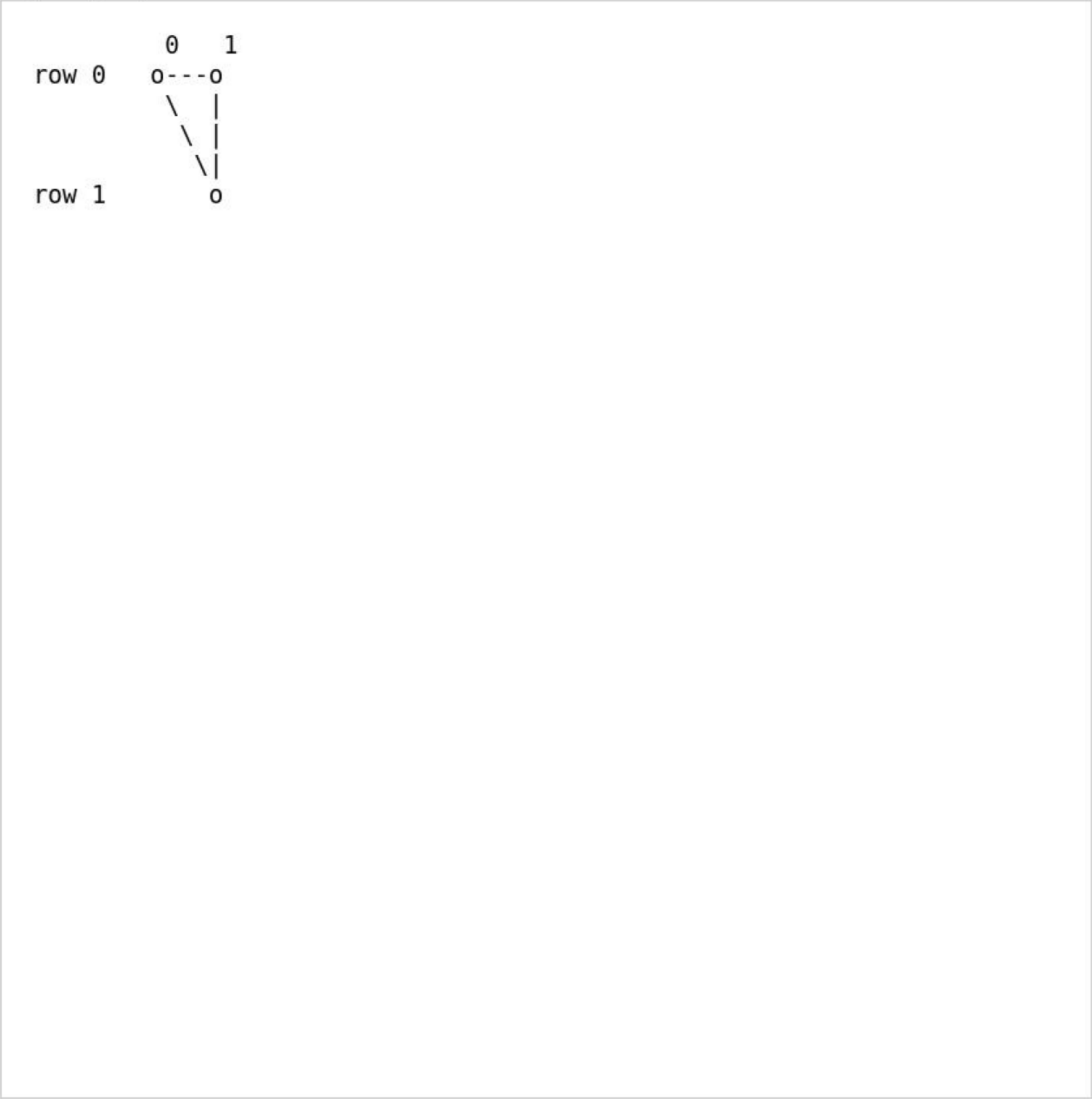
Output braid

Diagram illustrating a crossing between two strands, labeled 1, 2, 3, and 4 at both top and bottom. The crossing is represented by two dashed lines that cross each other. To the right of the diagram is the code `<- sigma_2^(-1)`.

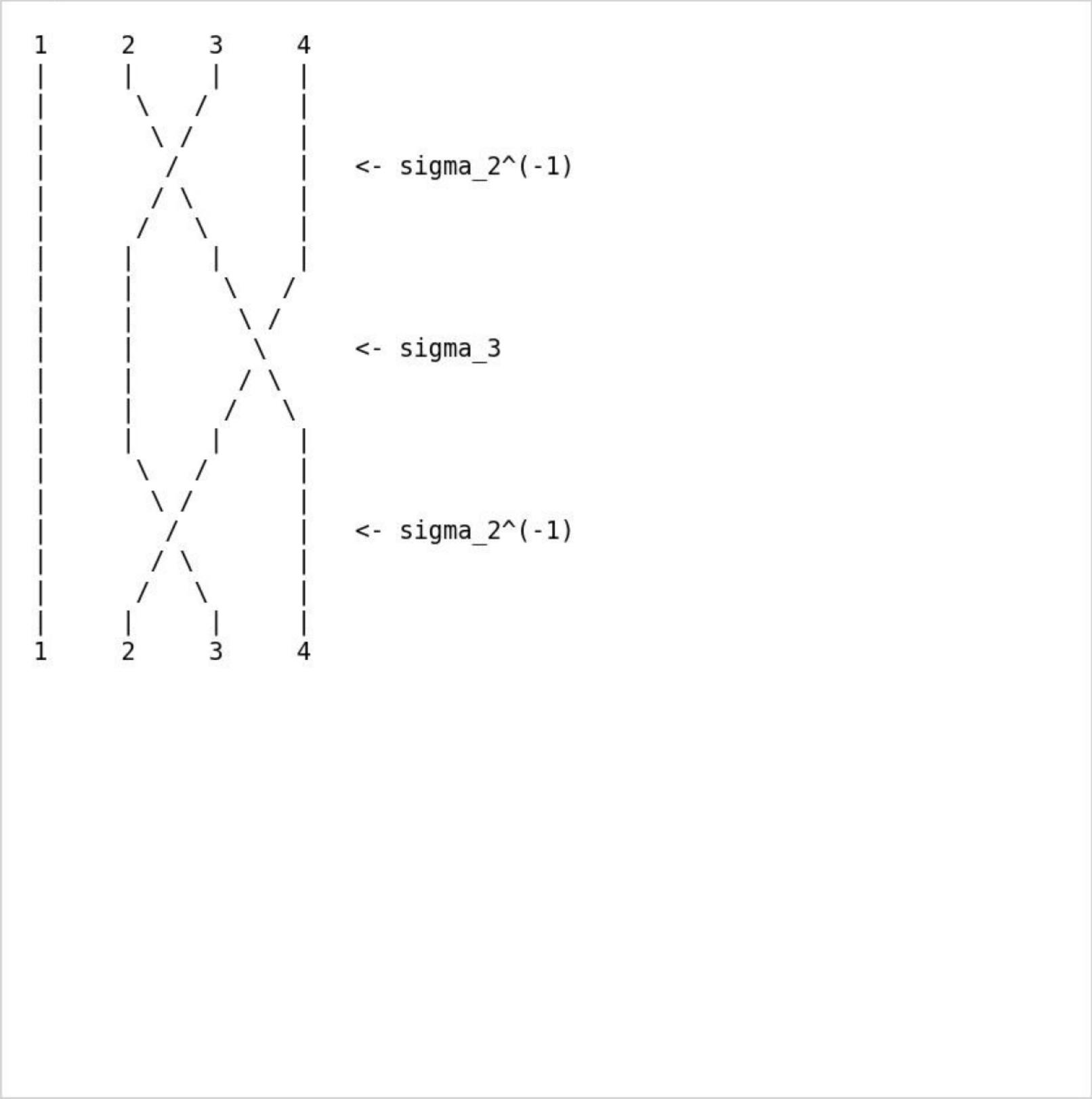
Example 8: Down-right diagonal — (0,0)-(1,1) plus horizontal (0,0)-(0,1)

Graph2Braid.py examples/example08_down_diagonal.json

Input graph



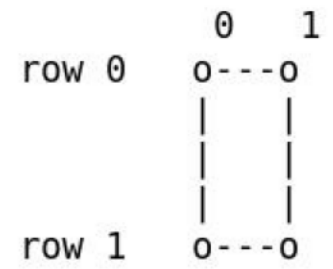
Output braid



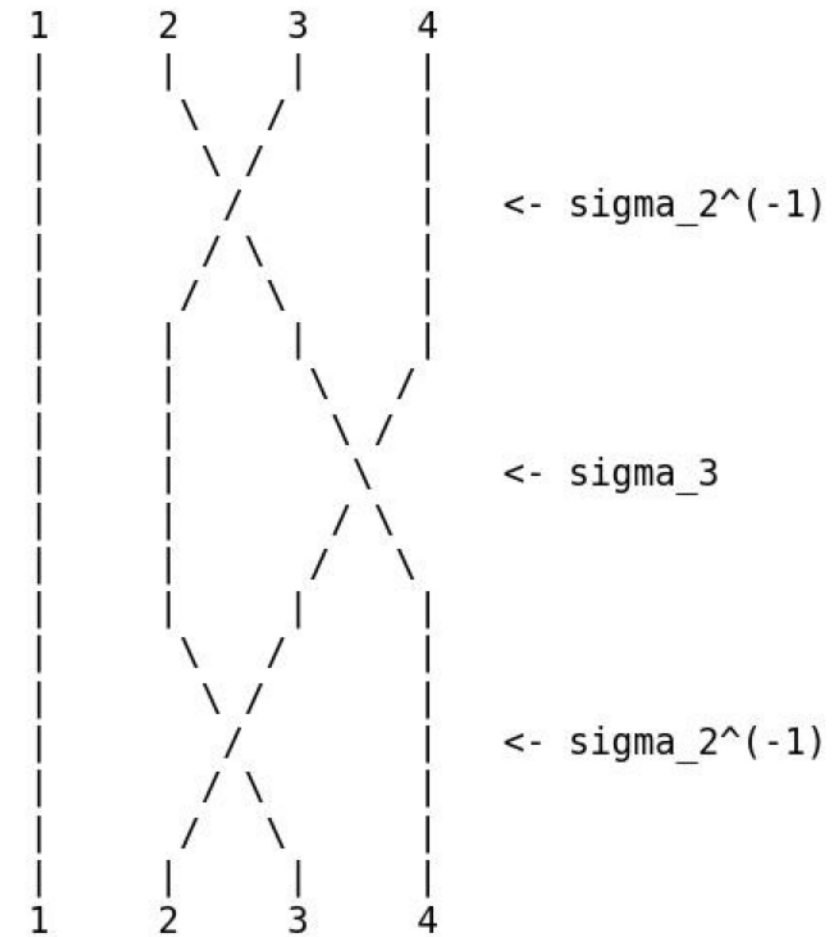
Example 9: Rectangle — same-row (horizontal) edges only

Graph2Braid.py examples/example09_same_row_edges.json

Input graph



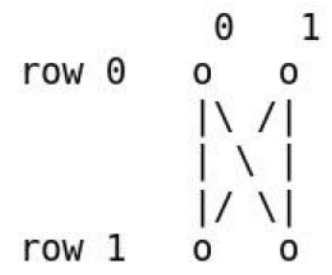
Output braid



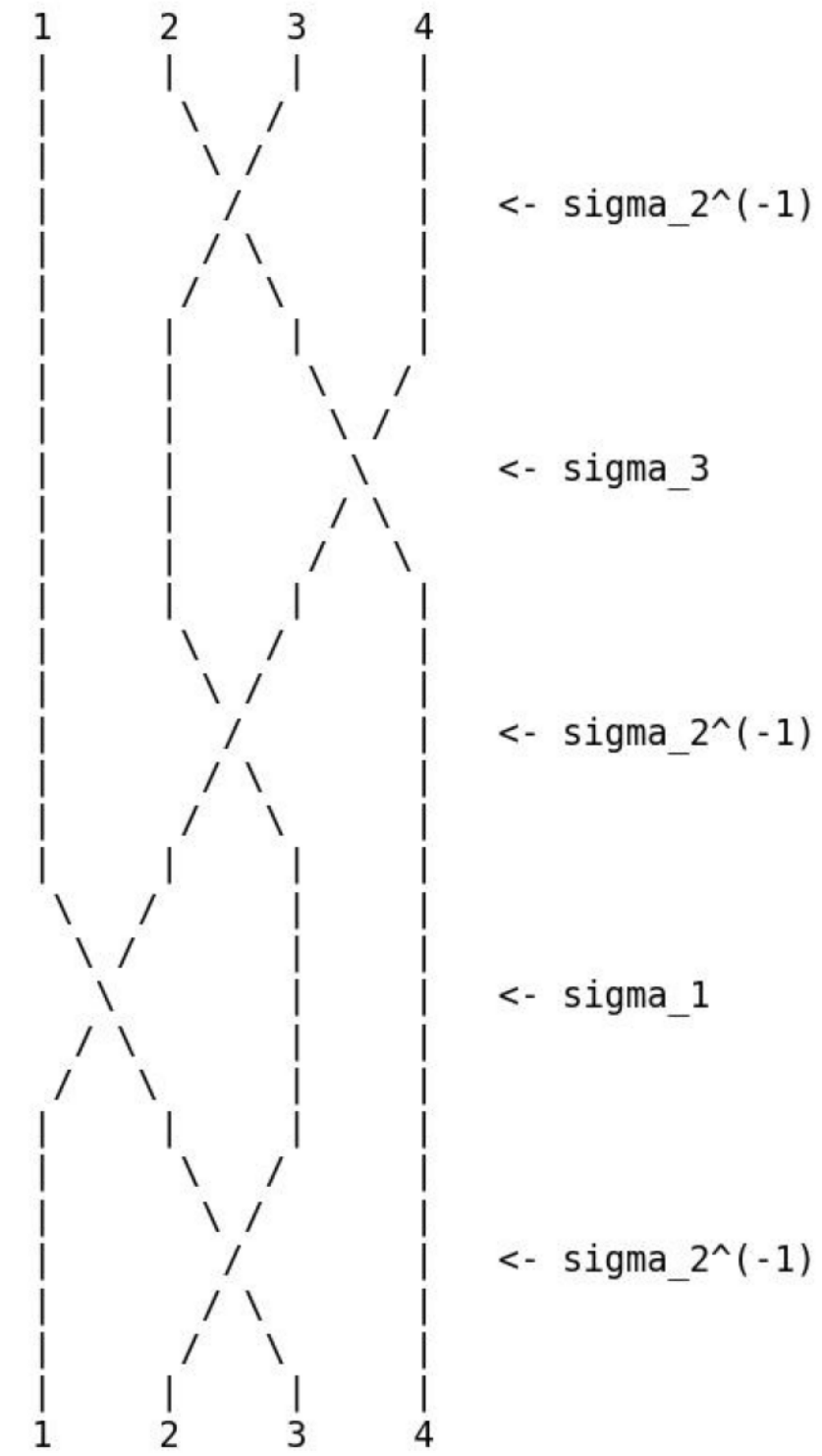
Example 10: Rectangle — cross diagonals only

Graph2Braid.py examples/example10_cross_diagonals.json

Input graph



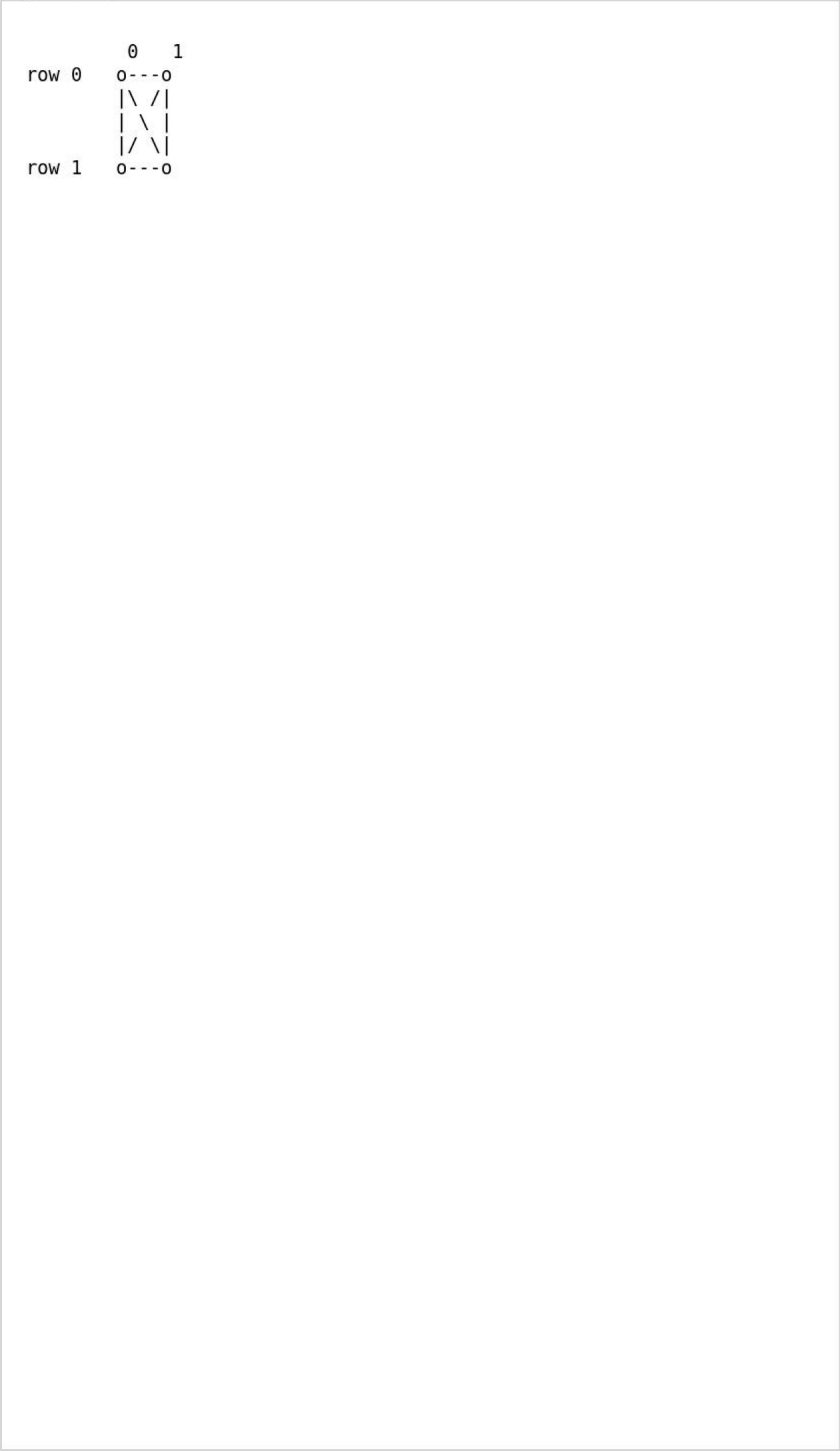
Output braid



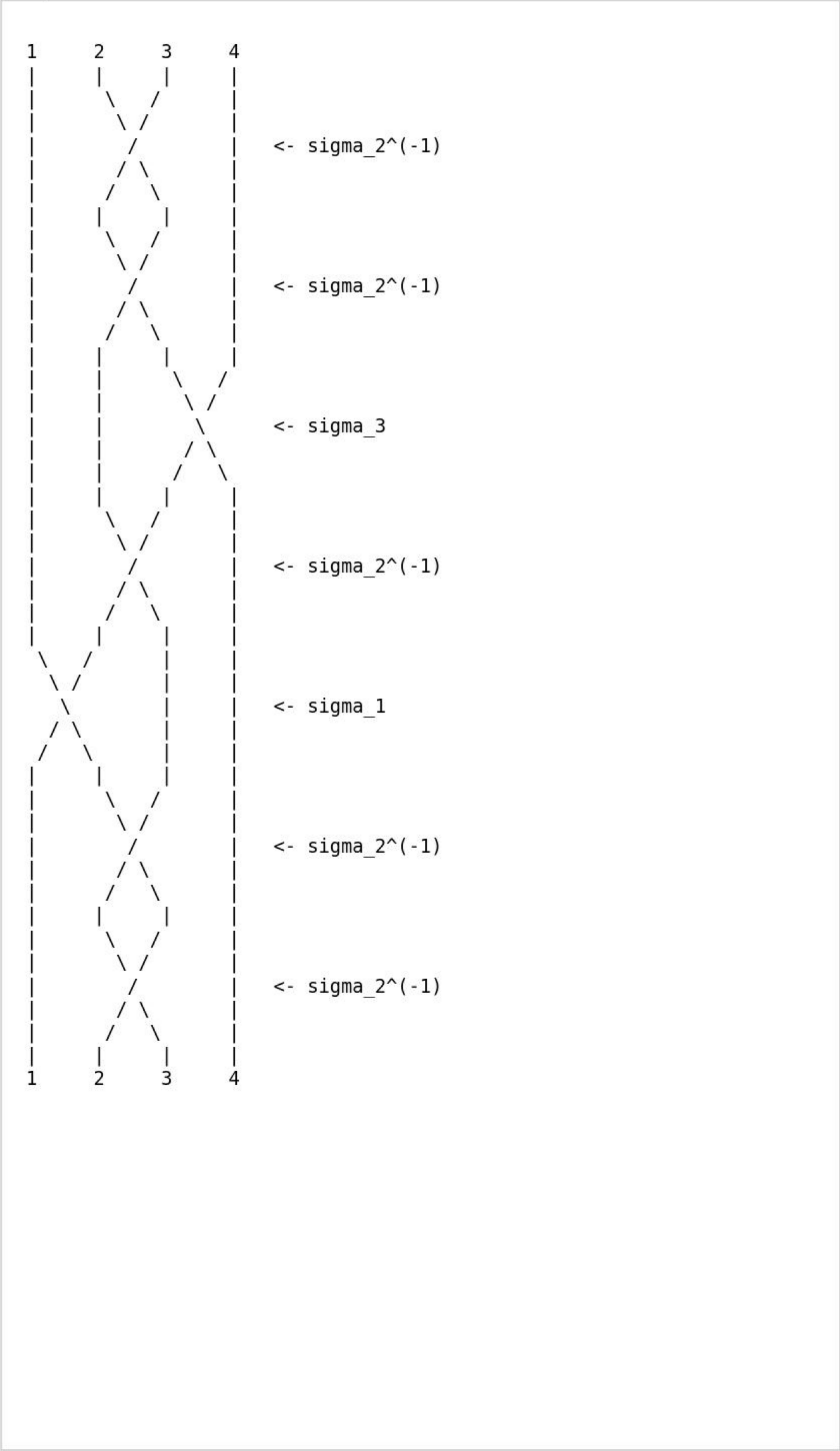
Example 11: Rectangle — all four non-vertical edges

Graph2Braid.py examples/example11_rectangle_all.json

Input graph



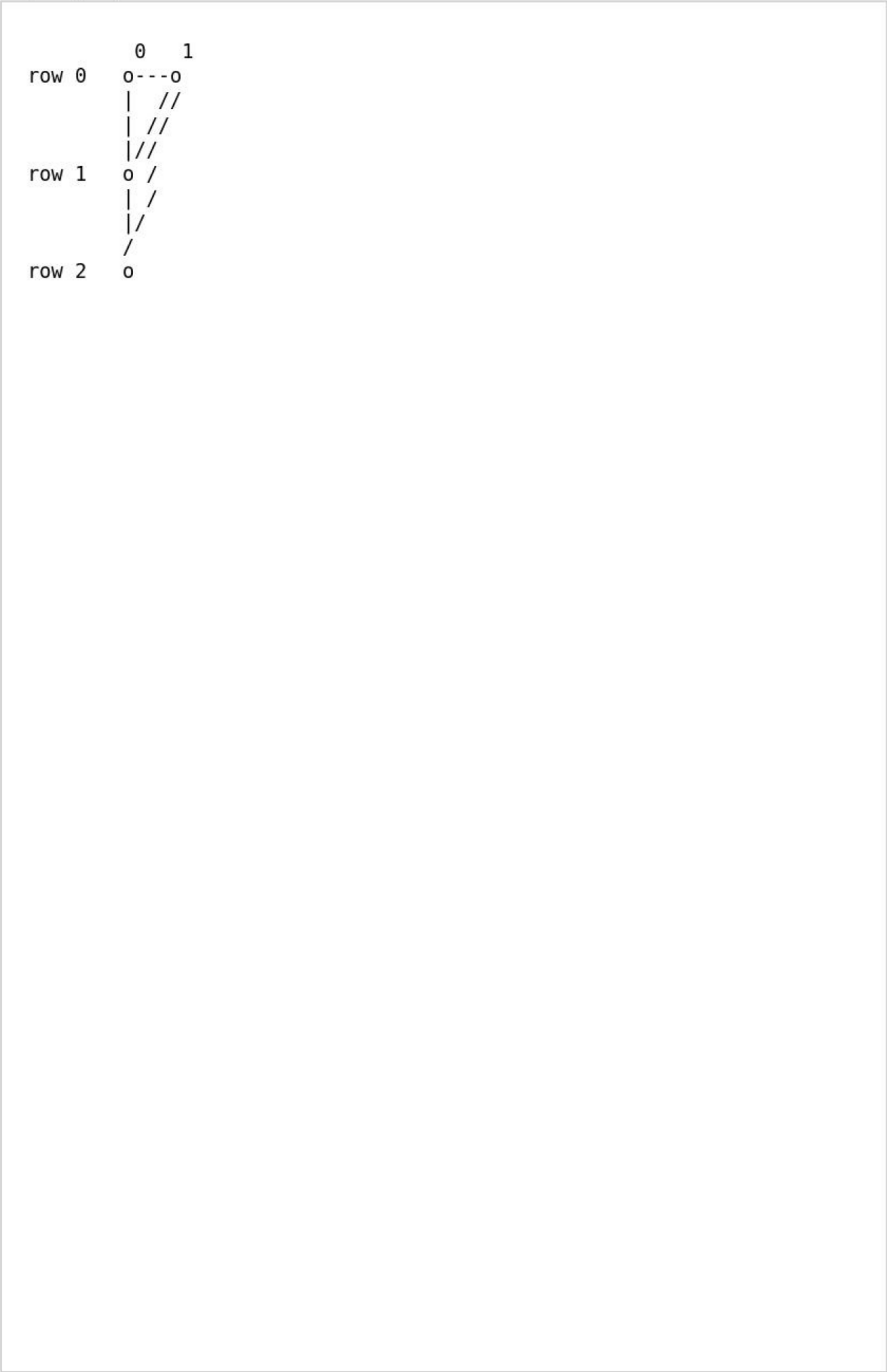
Output braid



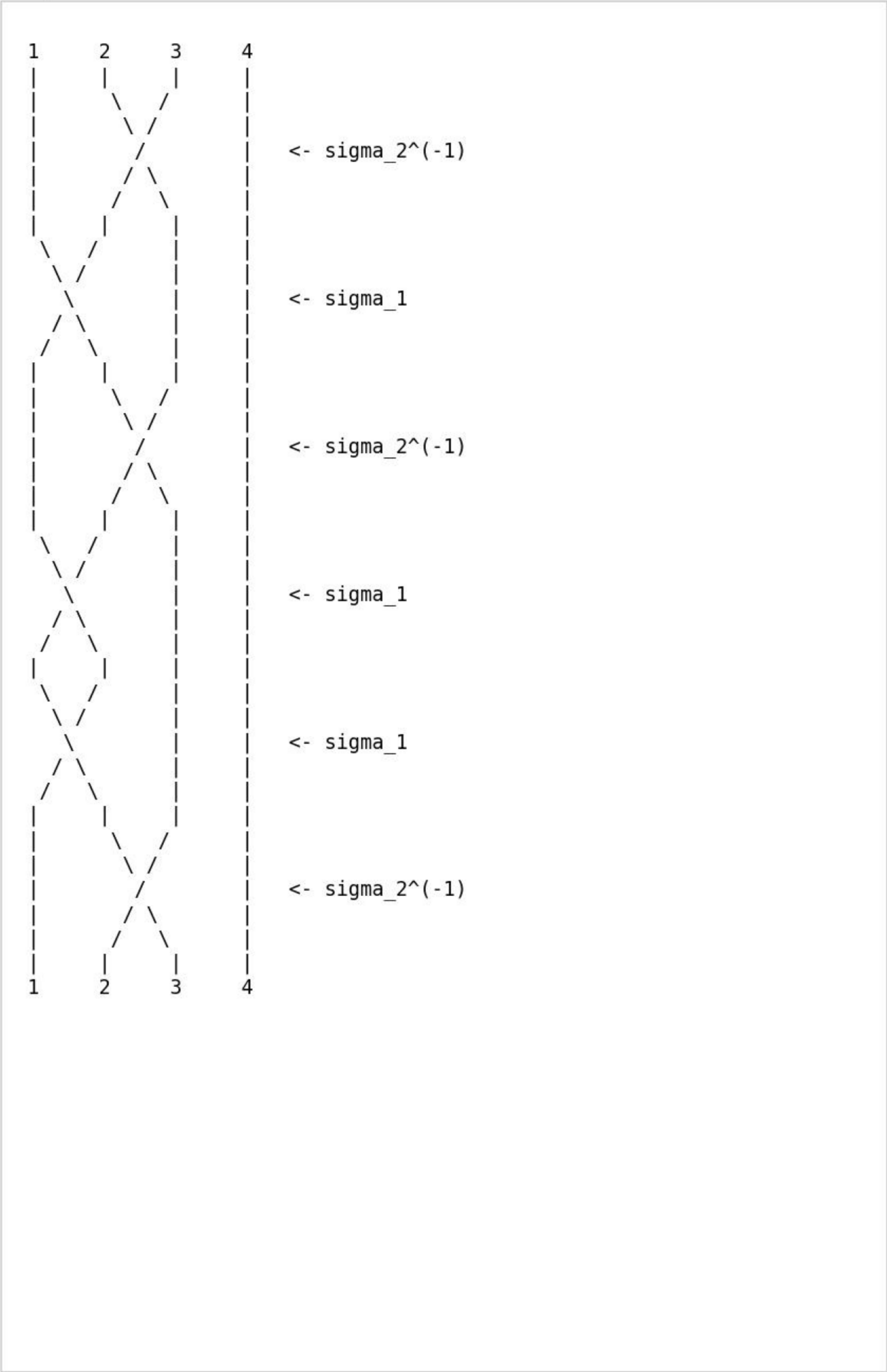
Example 12: Fan — three col-0 vertices all connected to (0,1)

Graph2Braid.py examples/example12_fan3.json

Input graph



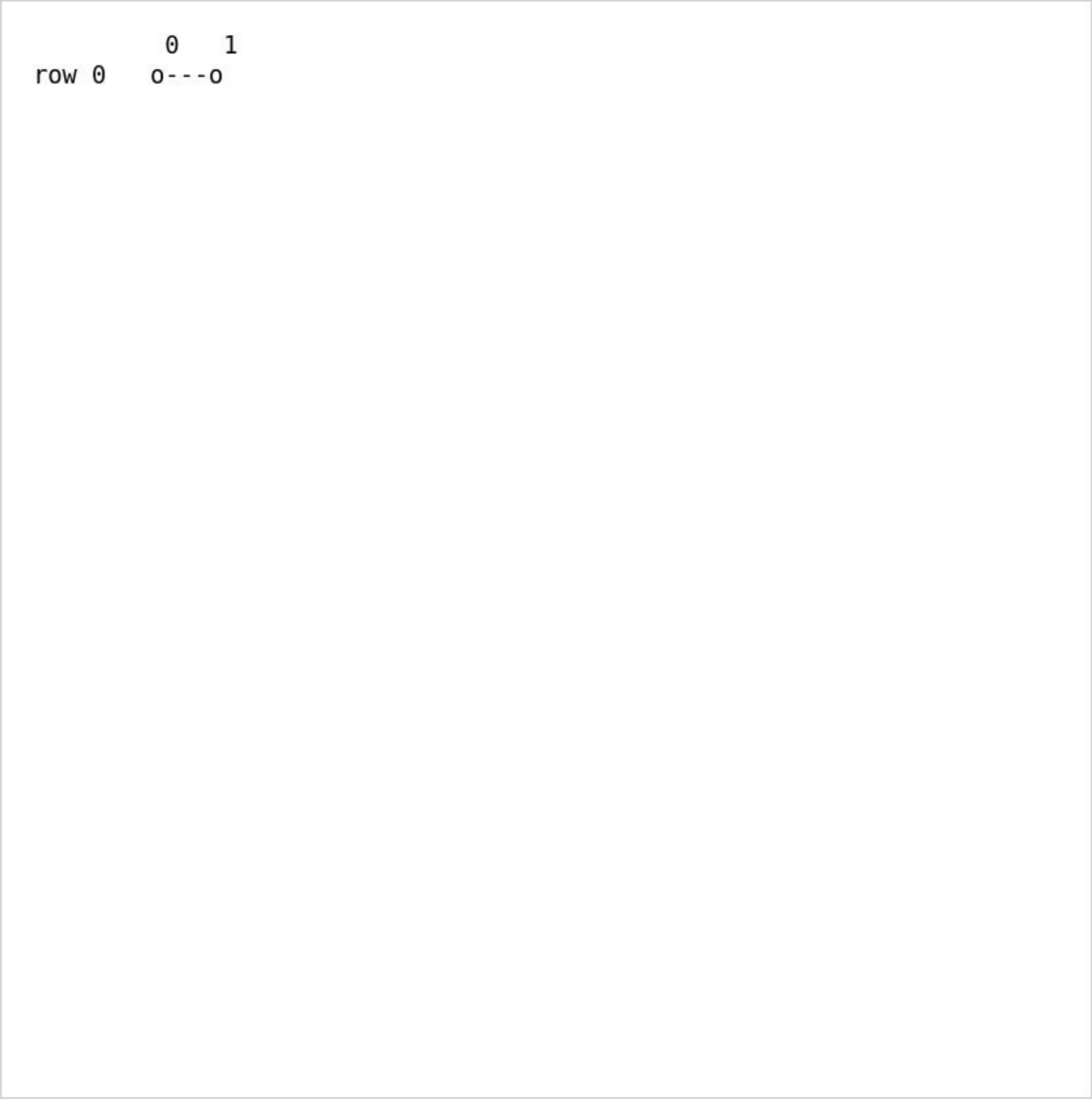
Output braid



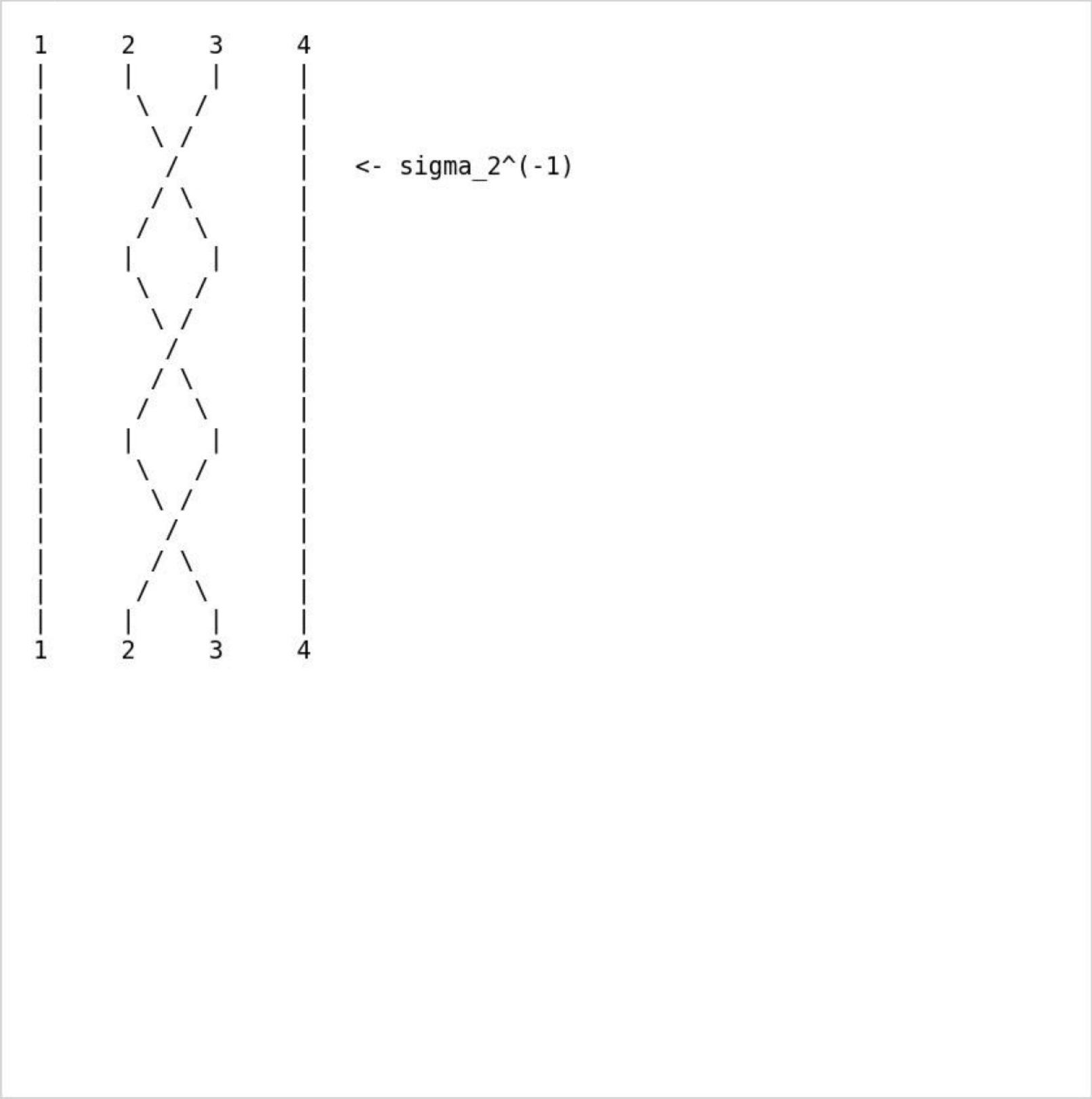
Example 13: Single edge with weight 3

Graph2Braid.py examples/example13_weight3.json

Input graph



Output braid



Example 14: Three vertices in row 0, two horizontal edges

Graph2Braid.py examples/example14_three_cols.json

Input graph

```

      0  1  2
row 0  0---0---0

```

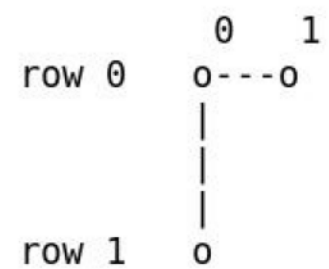
Output braid

Diagram illustrating a quantum circuit with 6 qubits. The circuit consists of two layers of gates. The first layer has a CNOT gate from qubit 2 to qubit 3, and a CNOT gate from qubit 4 to qubit 5. The second layer has a CNOT gate from qubit 3 to qubit 2, and a CNOT gate from qubit 5 to qubit 4. The input state is $|1\rangle$ on qubit 1, $|2\rangle$ on qubit 2, $|3\rangle$ on qubit 3, $|4\rangle$ on qubit 4, $|5\rangle$ on qubit 5, and $|6\rangle$ on qubit 6. The output state is $|1\rangle$ on qubit 1, $|2\rangle$ on qubit 2, $|3\rangle$ on qubit 3, $|4\rangle$ on qubit 4, $|5\rangle$ on qubit 5, and $|6\rangle$ on qubit 6.

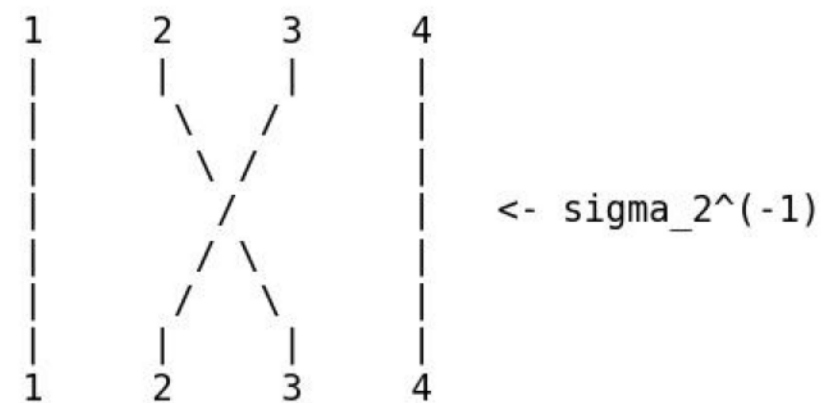
Example 15: L-shape with horizontal edge only — no diagonal

Graph2Braid.py examples/example15_horizontal_only.json

Input graph



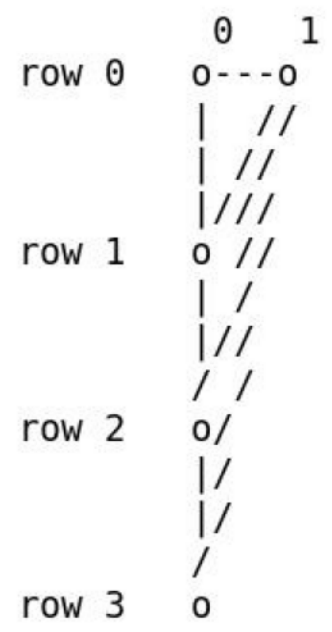
Output braid



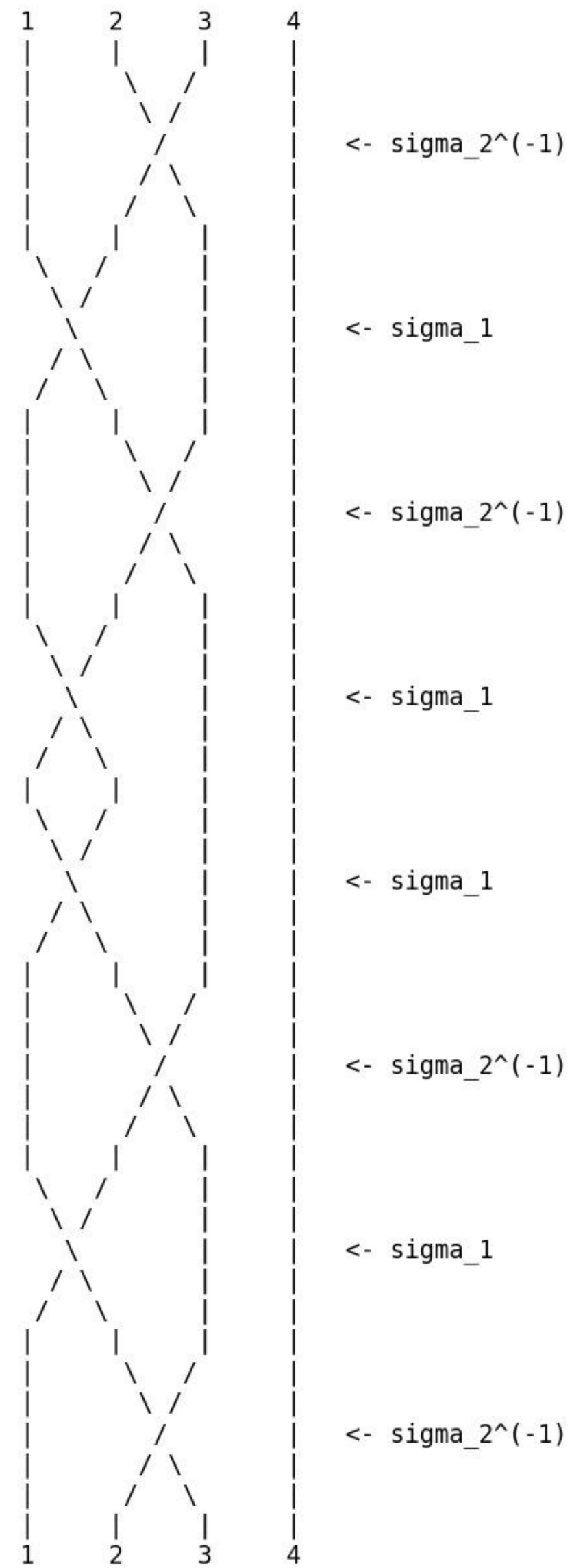
Example 16: Deep fan — four col-0 vertices all connected to (0,1)

Graph2Braid.py examples/example16_deep_fan4.json

Input graph



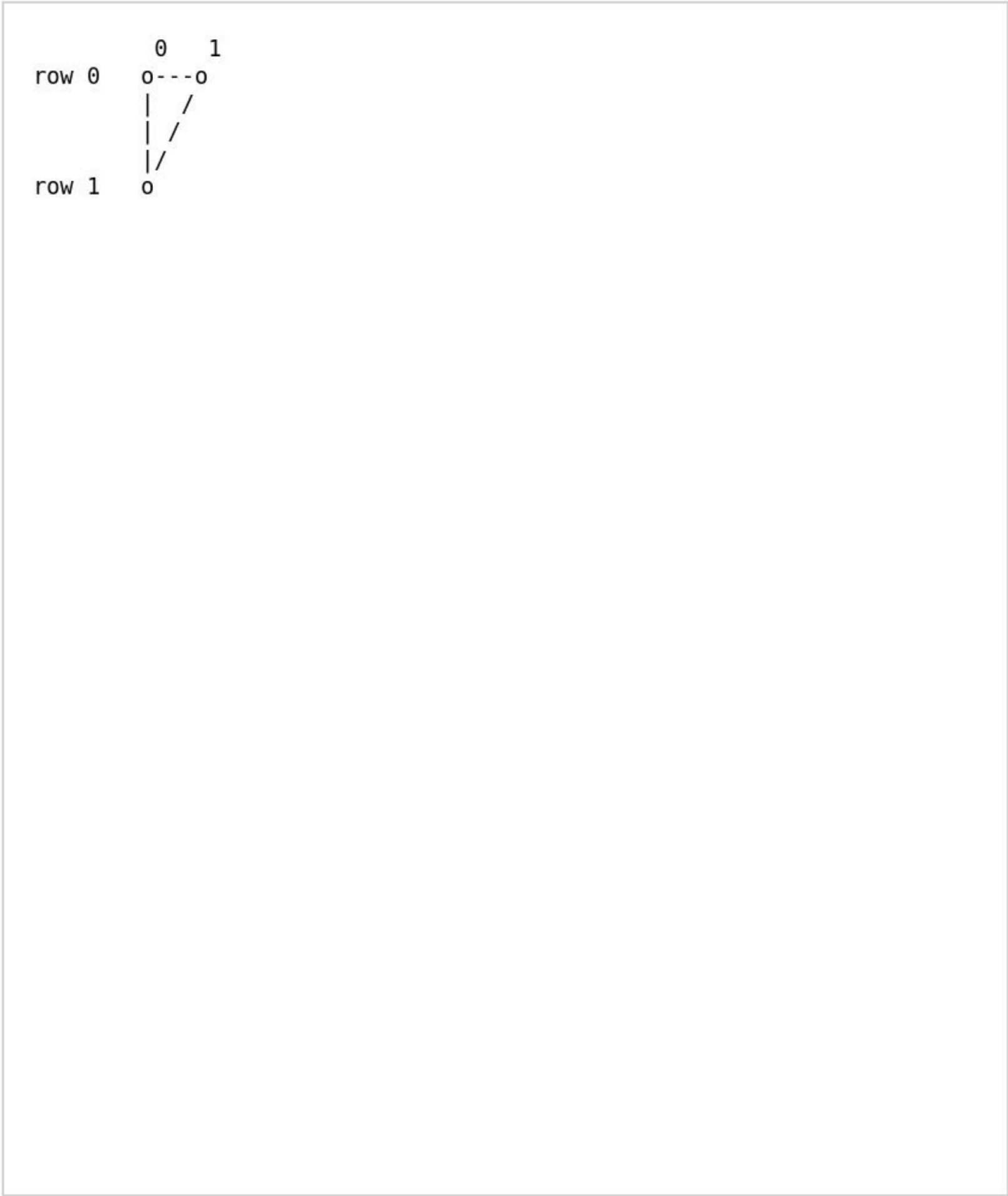
Output braid



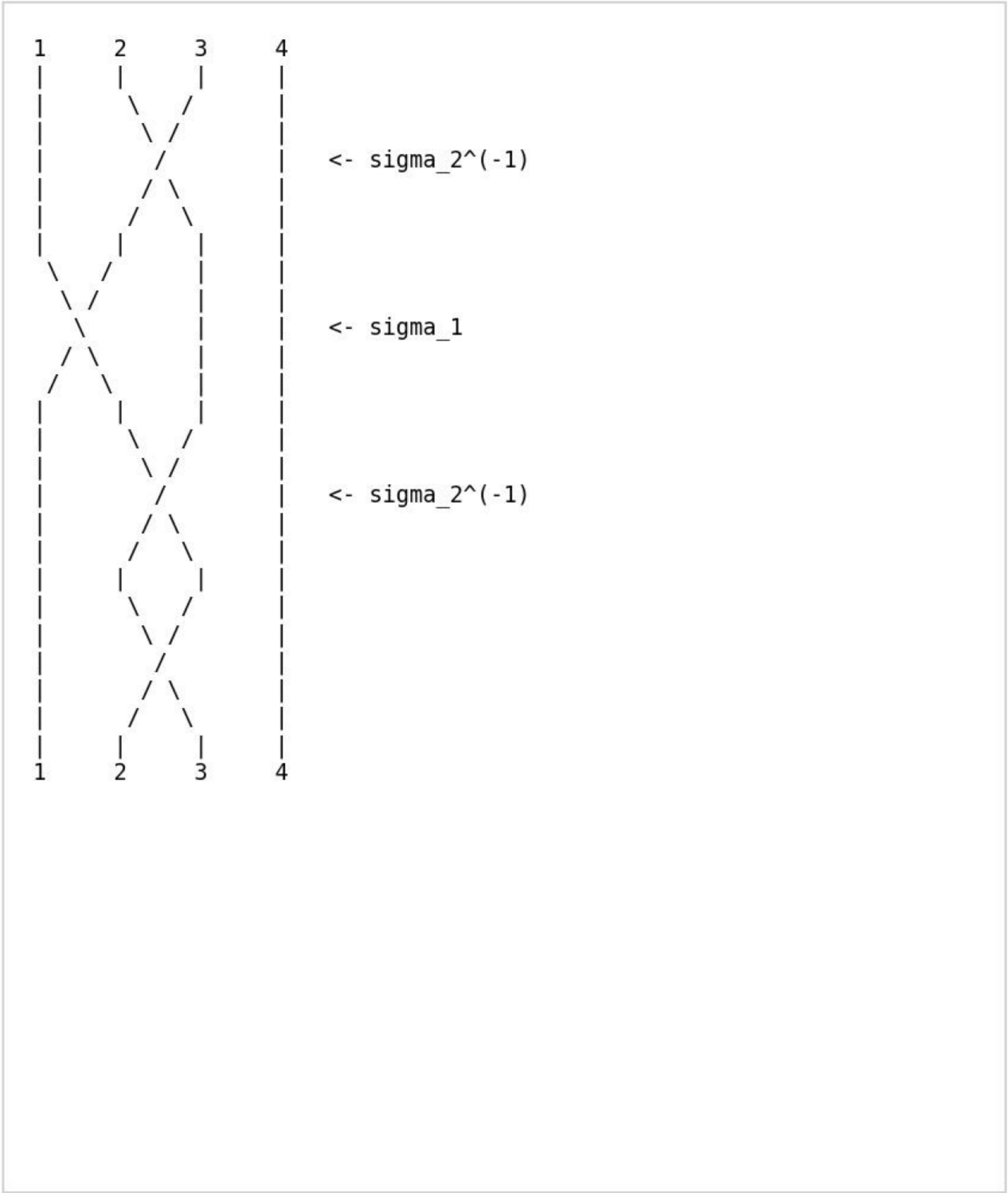
Example 17: Triangle with mixed weights — horizontal weight 1, diagonal weight 2

Graph2Braid.py examples/example17_mixed_weights.json

Input graph



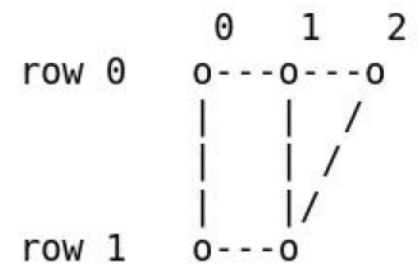
Output braid



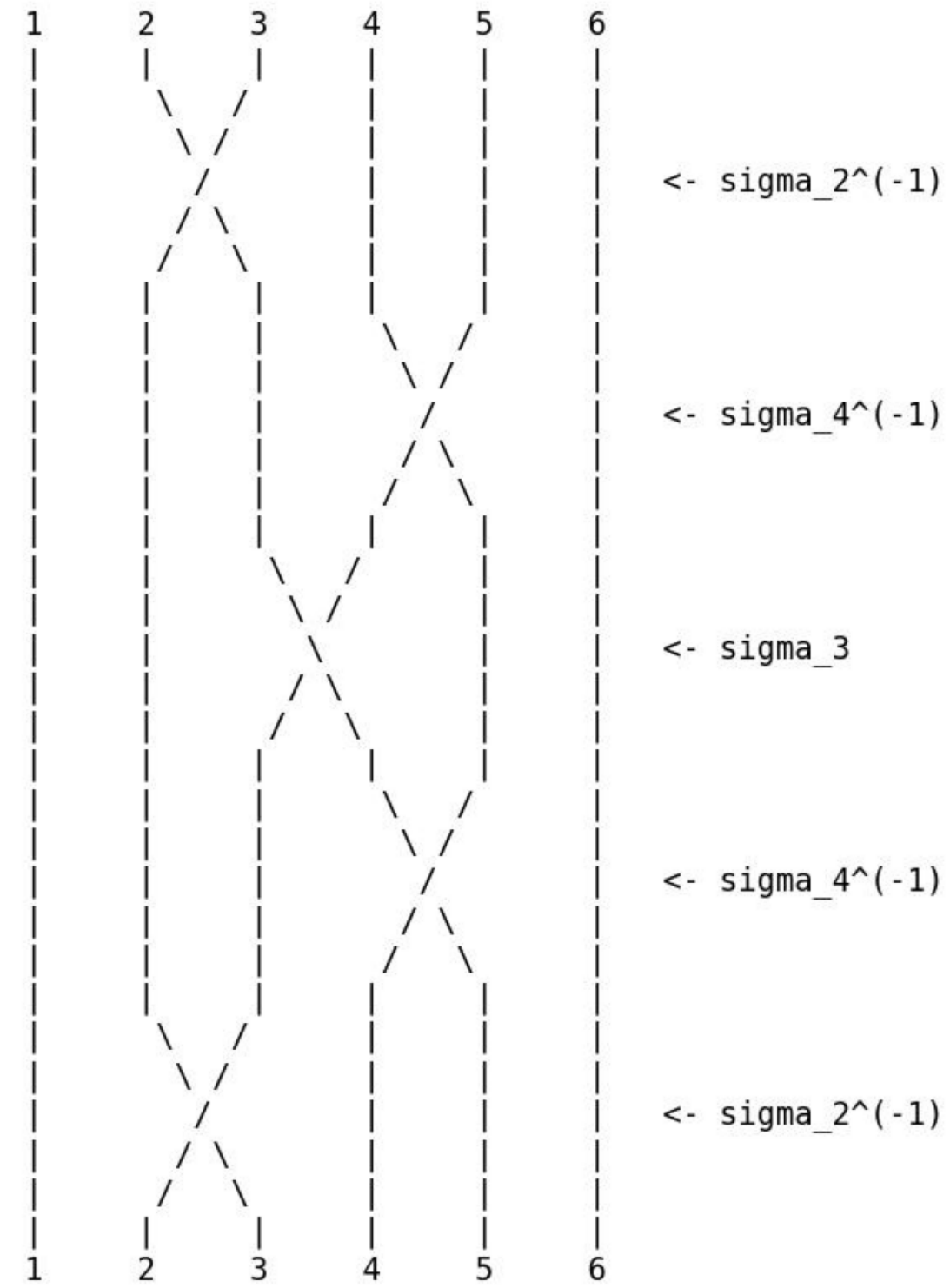
Example 18: Three-column strip — horizontal edges in rows 0 and 1

Graph2Braid.py examples/example18_strip.json

Input graph



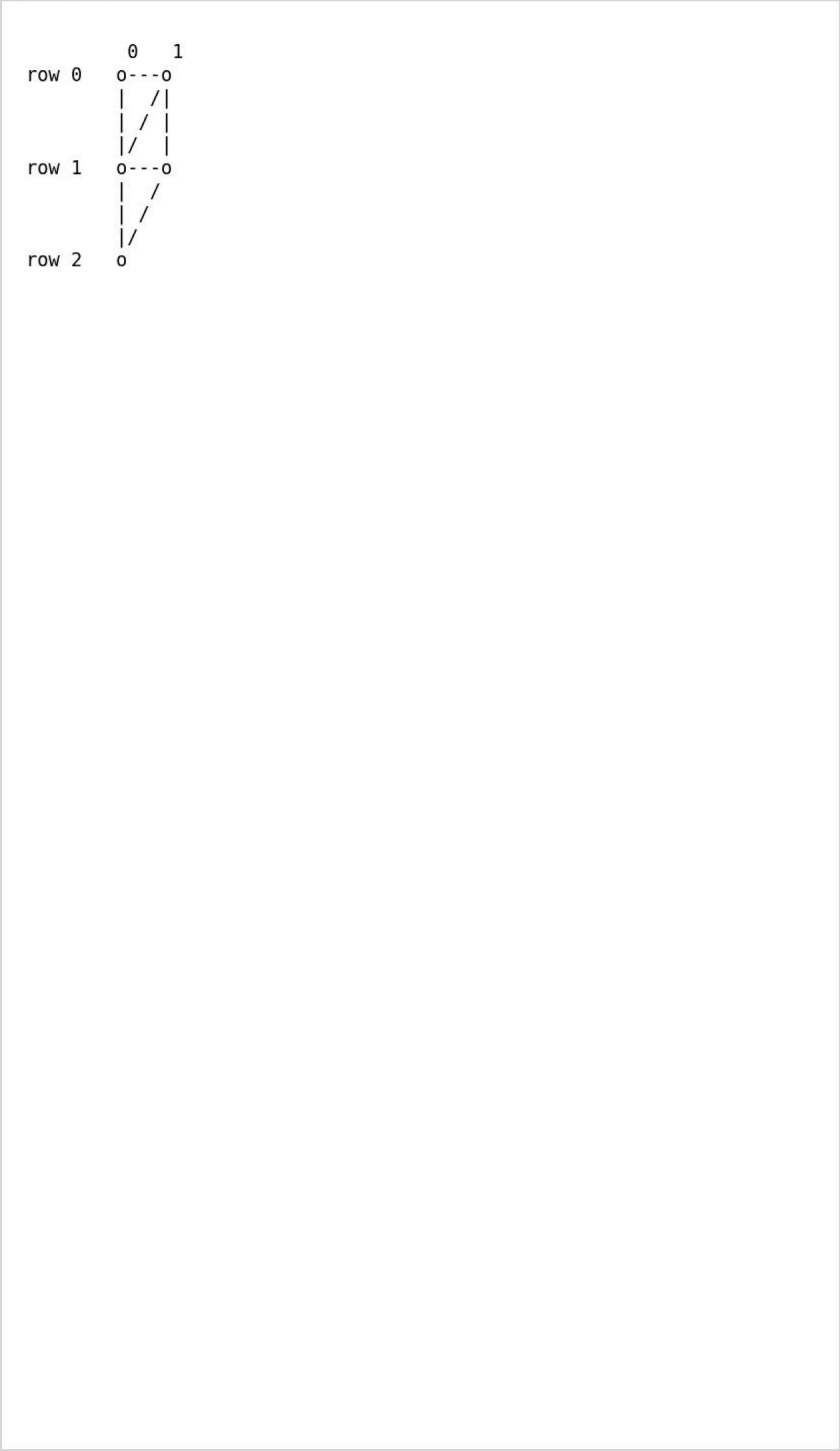
Output braid



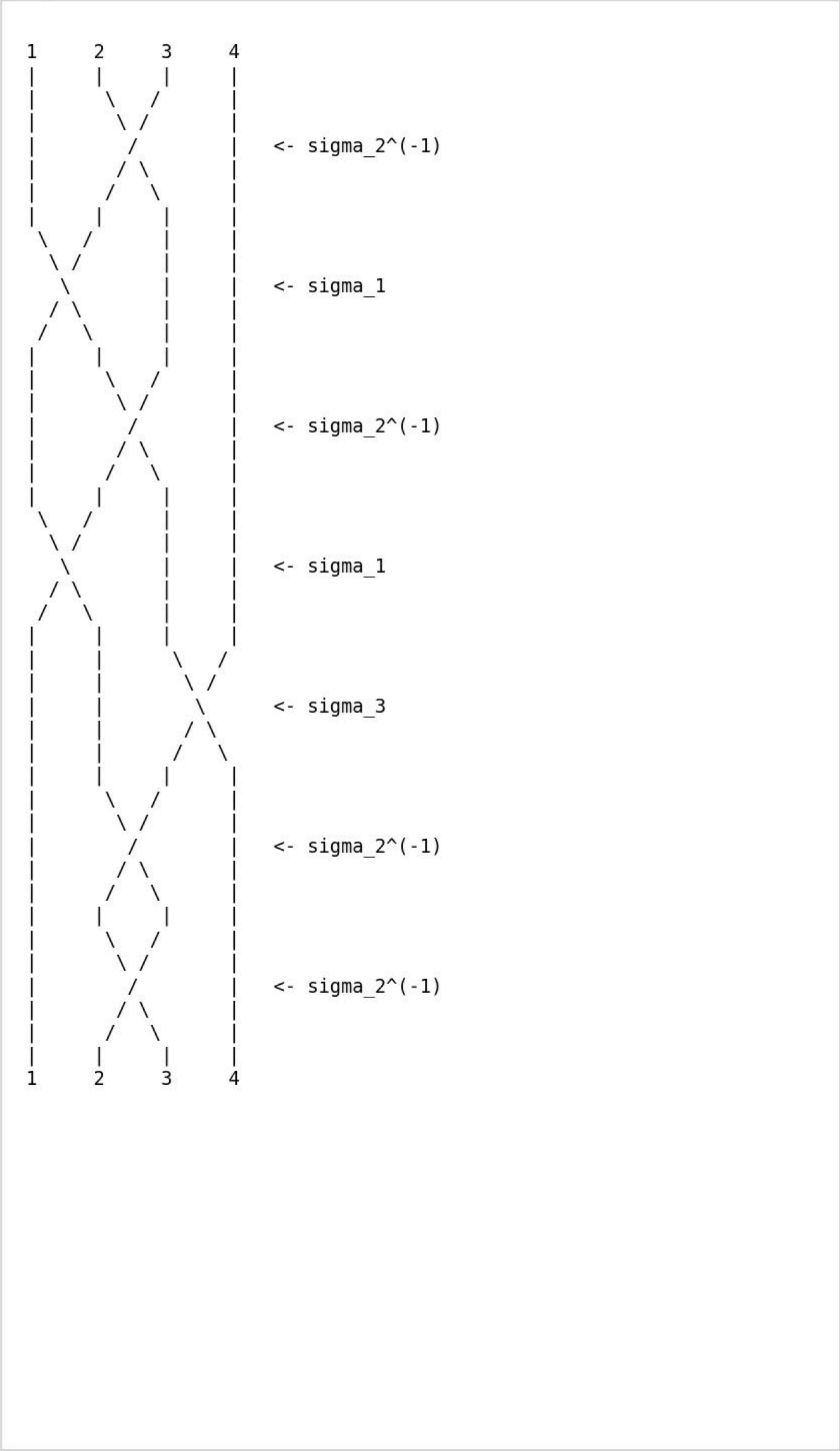
Example 19: Sub-staircase — lower-left portion of the 3x3 staircase

Graph2Braid.py examples/example19_substaircase.json

Input graph



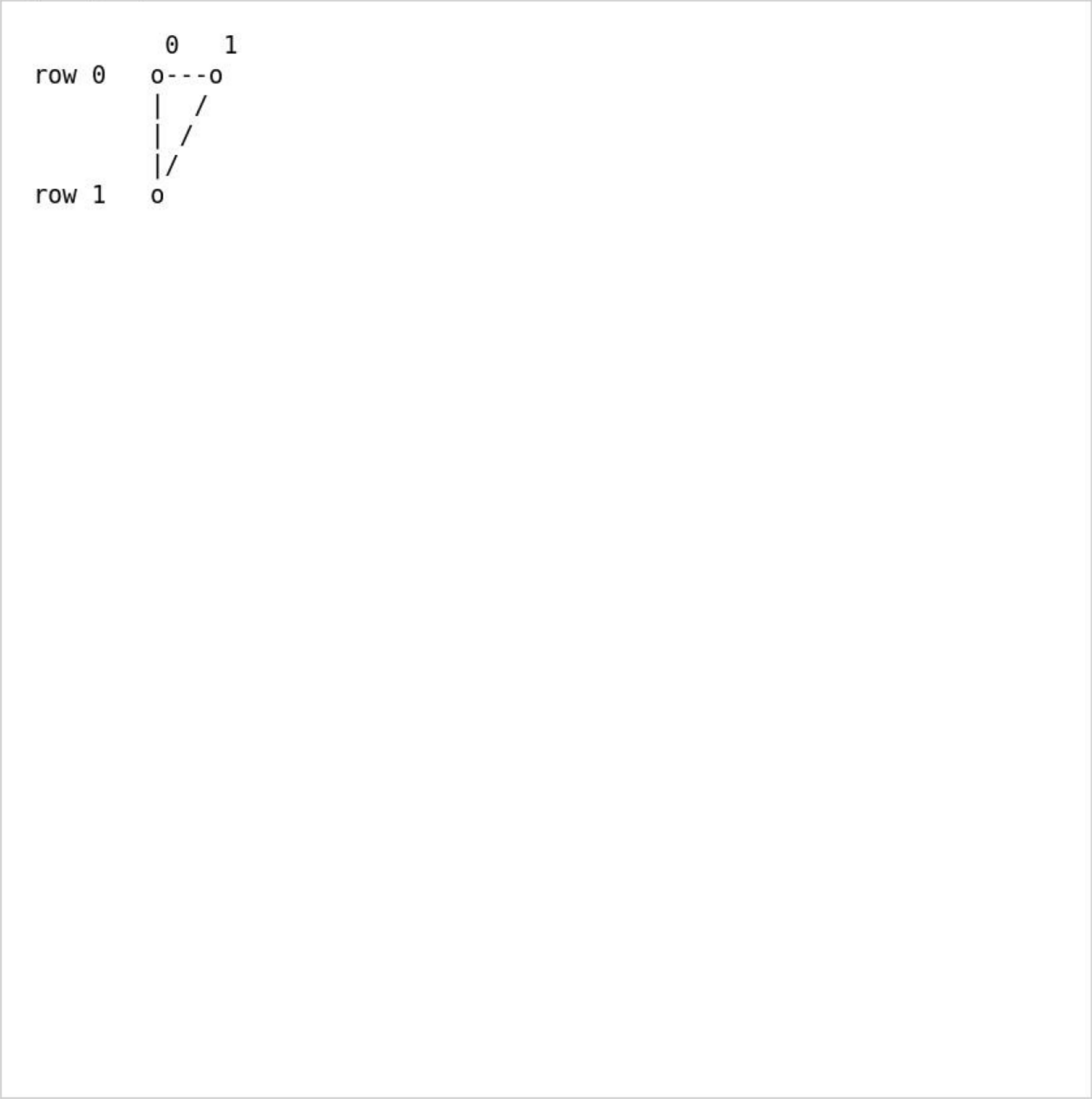
Output braid



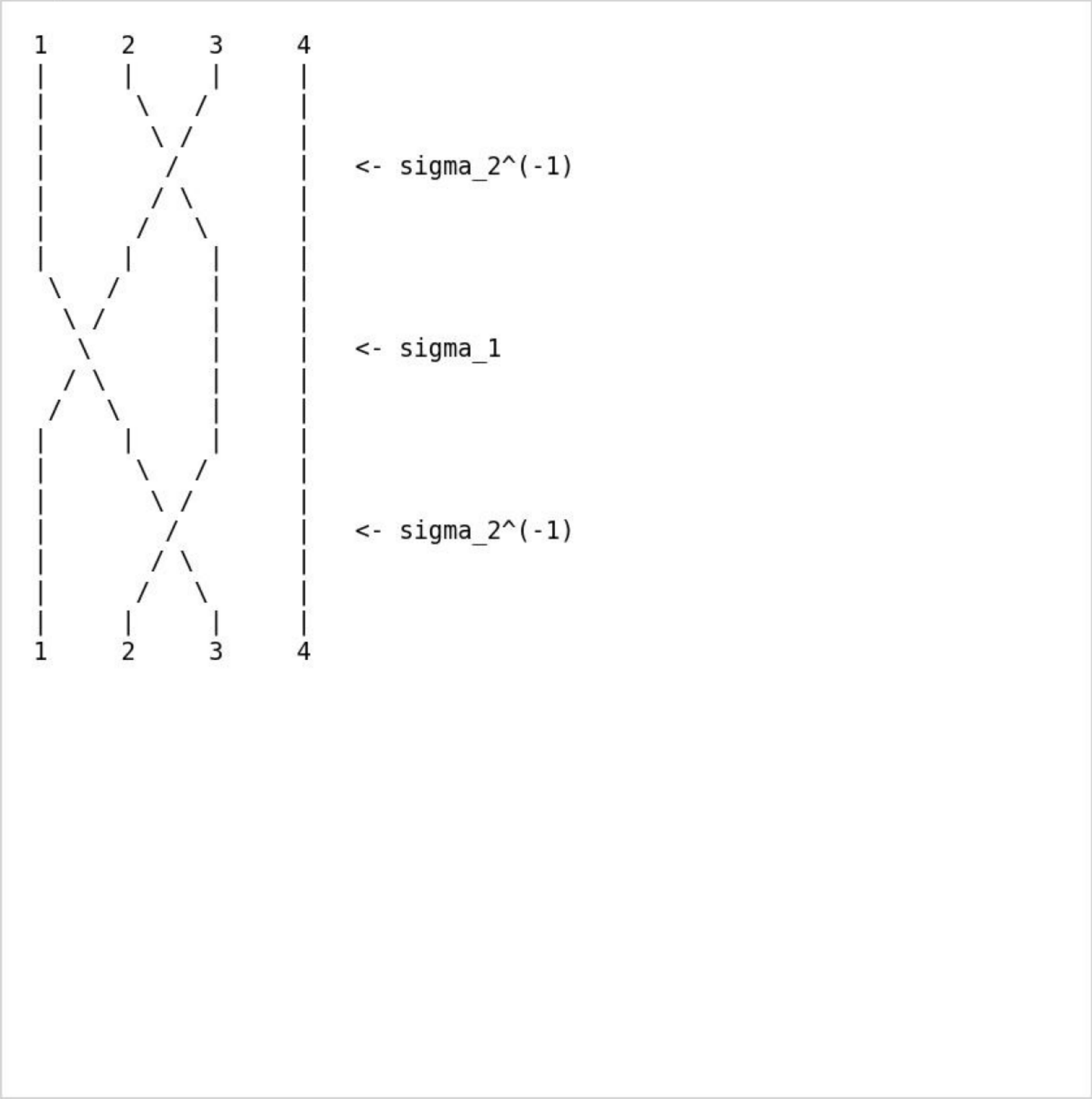
Example 1: Triangle — vertices (0,0), (0,1), (1,0)

Graph2Braid.py examples/example1_triangle.json

Input graph



Output braid



Example 2: Single non-vertical edge between (0,0) and (0,1)

Graph2Braid.py examples/example2_single_edge.json

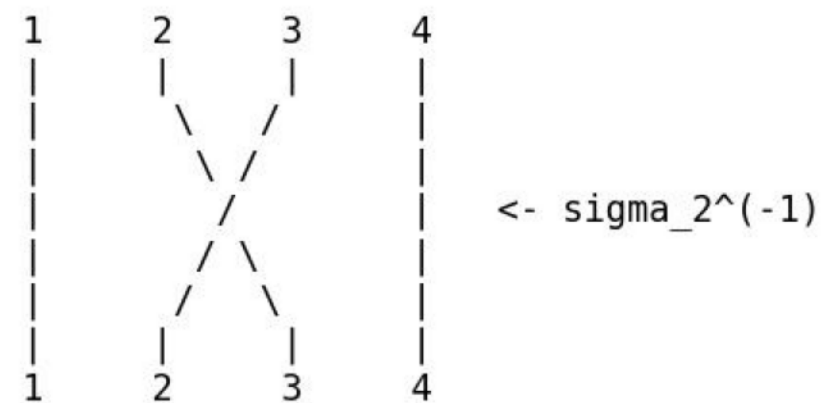
Input graph

```

      0  1
row 0  0 - - 0

```

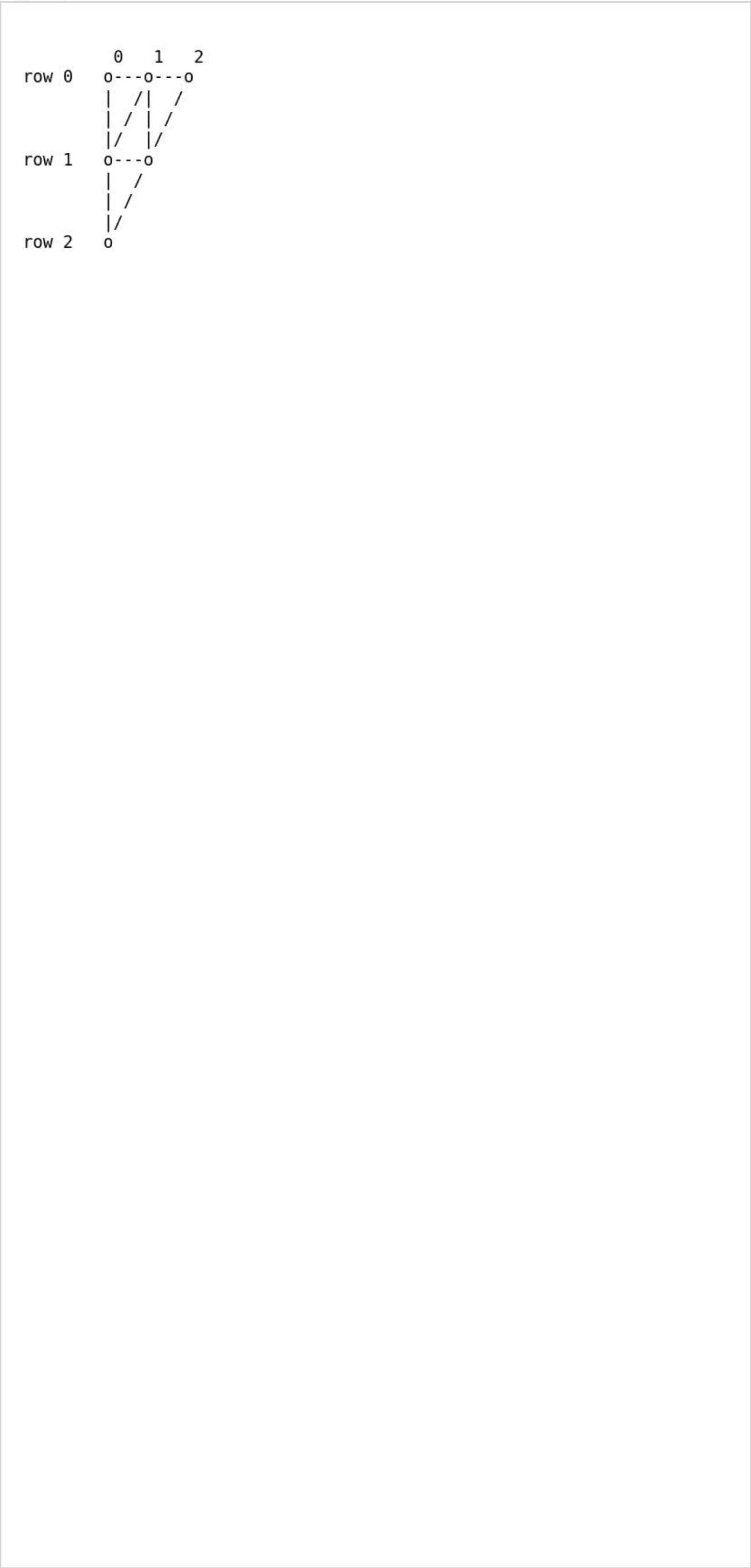
Output braid



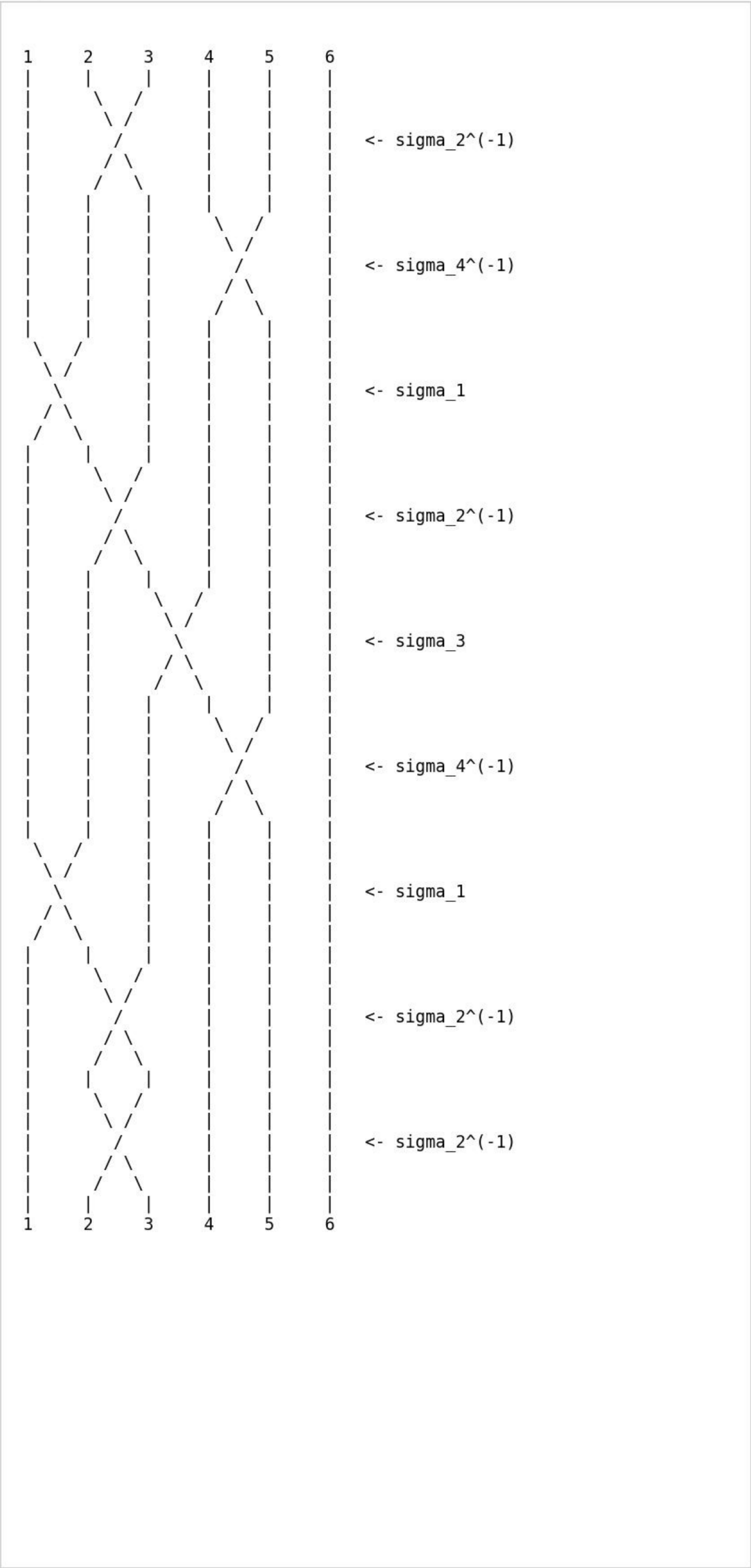
Example 3: 3-row staircase on a 3x3 grid

Graph2Braid.py examples/example3_staircase.json

Input graph



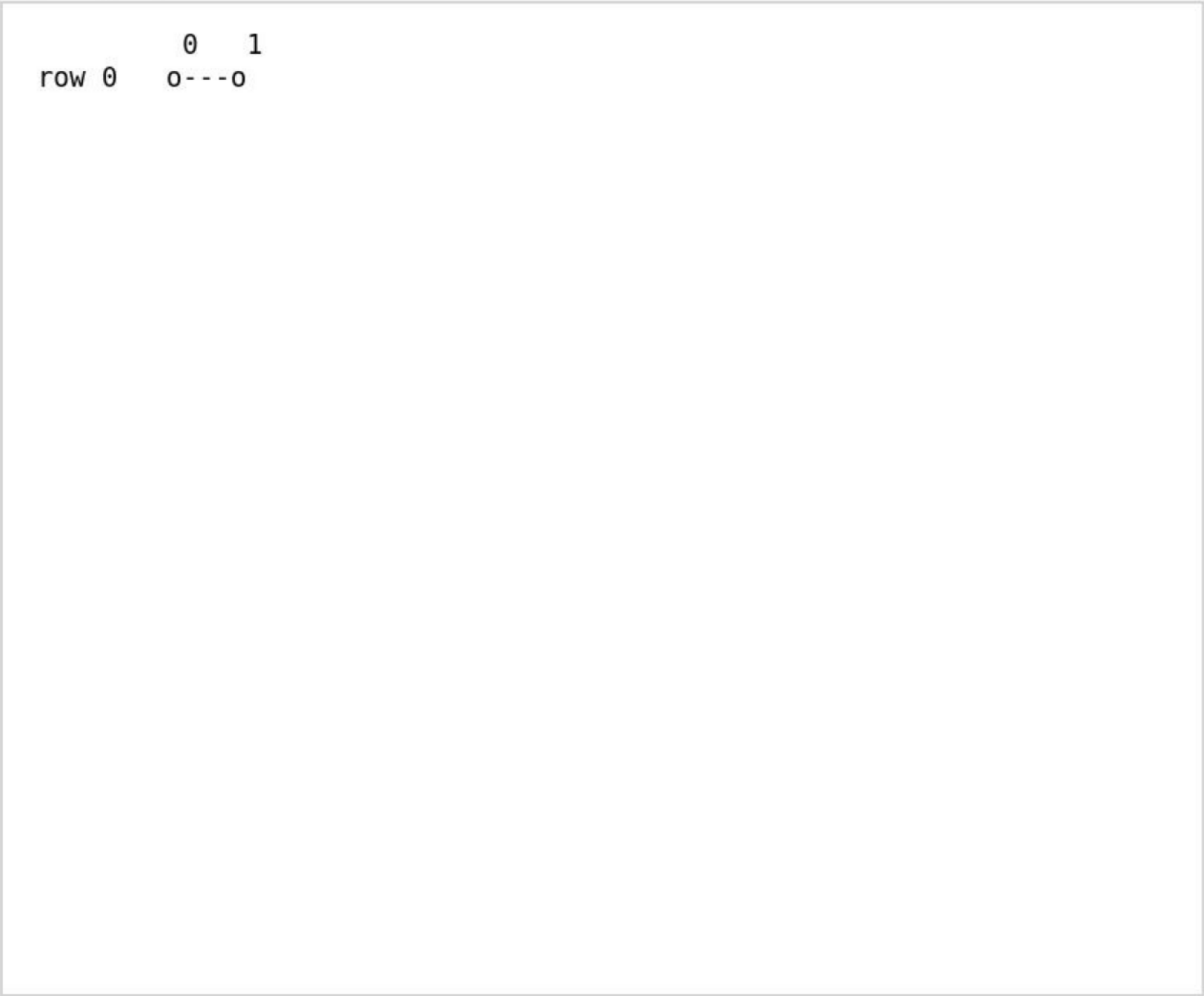
Output braid



Example 4: Single edge with weight 2 (two parallel edges between (0,0) and (0,1))

Graph2Braid.py examples/example4_double_edge.json

Input graph



Output braid

